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An intelligent HCI based tutoring system for child-centric learning environment

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Abstract
The scientific studies in computer based expert systems have improved in the area related to the disabilities identification and development of support systems for learning disabled people. Several disabilities have been identified with tools to help learners using various scientific methods. The children with learning difficulties have concerns in learning and capabilities to perform well in academics. An intelligent tutoring system (ITS) is an evolutionary computer expert system in the education domain that is primarily developed with the objective of teaching and learning process. In the present work, the ITS "Multisensory Instructions with Remedial Assistance in Child-centric Learning Environment" (MIRACLE) is developed utilizing machine learning models to identify learning disabilities, such as Dyslexia, Dysgraphia, and Dyscalculia. The MIRACLE performs identification and presents learner-specific content to the learners using machine learning methods. A total of 304 participants learning disabled and non-learning disabled were observed to carry out the experiment using a pretest and some non-invasive tools. The framework MIRACLE for the learning disabilities identification is implemented as a hybrid model with LSTM combined with SVM and Analytic Hierarchy Process. The results indicate the efficacy of the proposed ITS for the learning disabled.

Keywords Special Education · Classification; Learning Disabilities; Analytical Hierarchical Process · Intelligent Tutoring System

Abbreviations

AI	Artificial Intelligence
MIRACLE	Multisensory Instructions with Remedial Assistance in Child-centric Learning Environment
ITS	Intelligent Tutoring System
LDs	Learning difficulties

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RNN	Recurrent Neural Network
SVM	Support Vector Machine
LSTM	Long-short term memory
AHP	Analytical Hierarchical Process
SLD	Specific Learning Disability
IDEA	Individuals with Disabilities Education Act
ANN	Artificial Neural Network

1 Introduction

The Individuals with Disabilities Education Act (IDEA) uses the term Specific Learning Disability (SLD) to describe learning disabilities (LD). It defines SLD as a disorder affecting language, speaking, and writing abilities, rooted in fundamental psychological processes related to understanding and language skills. This category encompasses cognitive disabilities, dyslexia, dysgraphia, perceptual disabilities, and dyscalculia [1].

Developmental Dyslexia, affecting learning, speaking, and writing skills, impacts approximately 5–12% of individuals, often leading to difficulties in word processing and decoding [2–4]. Dysgraphia symptoms include challenges with motor skills, spatial knowledge, and cognitive abilities. Dyscalculia in children manifests as difficulty recognizing numbers and understanding arithmetic operations [5–7]. These challenges significantly affect children's academic performance and self-esteem among peers. Individuals with Dyslexia typically possess normal intelligence but struggle with delayed language processing, impacting their writing, speaking, and comprehension skills [8]. Early identification of learning difficulties is crucial for effective intervention and support by psychologists, specialists, and therapists. In education, the rise of adaptive e-learning environments allows for personalized learning experiences tailored to individual learner preferences [9]. To optimize learning outcomes, these environments must be finely tuned to accommodate the specific characteristics and needs of each learner. To effectively manage the profiling of learners with learning disabilities, it is essential to design a more efficient learner identification model for profiling. Literature shows that hybrid learning models or combining multiple learning models significantly improve the performance of individual classification or clustering models. Typically, the first stage preprocesses the learning data, while the second stage handles the final classification task [10].

In this study, we explore two methods for cascading different learning models to identify learning disabilities. Our cascading model integrates Long Short-Term Memory (LSTM) classification, Support Vector Machine (SVM) classification, and the Analytic Hierarchy Process (AHP) mathematical model (LSTM + SVM + AHP).

The objectives of this work are:

- To design and develop an Intelligent Tutoring System (ITS) for predicting learning disabilities, profiling learners, and generating personalized content.
- To evaluate the efficacy of the proposed ITS.

The paper is organized as follows: Sect. 2 reviews the literature on learning disability identification and learning models. Section 3 describes the ITS (MIRACLE) framework and the methodology used in its development. Section 4 discusses the experimental results, and Sect. 5 provides the conclusion.

2 Related Studies

2.1 The methodologies used for Learning Disabilities (LDs) prediction

The state-of-the-art techniques in Artificial Intelligence (AI) have advanced in developing computer-based technologies for supporting the learning disabled [10]. In some studies, EEG devices were used to collect appropriate brain regions for classification utilising machine learning [11–13]. The involvement of children in games and games related interfaces are also used for data collection in the Dyslexia identification process [14]. The learners with Dyslexia disability have a slower response to visual and auditory spatial attention than non-Dyslexic learners' visual and auditory capabilities [15]. A mobile application has monitored children's behavioral management and psychiatric assessment with ADHD symptoms. It makes the parents aware of the child-specific details and information [16]. In a study, ANN is implemented for unsupervised learning called the Growing Neural Forest to analyse the use of drugs MTF and melatonin for sleep disorders as sleep disorder is one of the comorbidities with other difficulties [17]. Neuromotor impairments like Dyspraxia have been solved using virtual reality environments with wearable haptics. The system usability has experimented on children with cerebral palsy (CP) and developmental dysgraphia (DD). Moreover, a kinematic assessment has also been designed in the system. Virtual Reality (VR) computer games have been used previously in some studies for identifying Dyspraxia to help occupational therapists [18, 19]. For dysgraphia identification, the software application TestGraphia is deployed for making the diagnosis and process easier for doctors. It is based on a document analysis algorithm [5].

2.2 The Long Short Term Memory (LSTM) for time-series data

The important information in Chinese text cannot be extracted easily because of Chinese language characteristics. The Hierarchical Comprehensive Context Modeling Network (HCCMN) is proposed in a study to extract this information from text hierarchically using the LSTM technique [20]. The text sentiment analysis approach is used to provide sentiment tendency of students to the selection committee for teaching evaluation of faculty members. It has also been analysed that the performance of the time series classifier is more effective when compared to the performance of the classifier not based on time series prediction [21]. A novel architecture is applied to discriminate representation of sentences for children's entity attribute extractions using the Class Game Representation (CGR). It transforms miRNA sequence information into image information and extracts its features. CGMDA combines miRNA sequence and miRNA functionality into disease semantic information to improve prediction accuracy, and finally, the classification of extracted data is classified using lightGBM methodology [22].

2.3 The existing frameworks and architectures

The methods that have been already utilised for learner disability identification and are the basis of our present work are discussed here. The methods that have been implemented in LDs prediction are machine or deep learning-based models. The SVM has proved to be an effective algorithm in most studies to predict learning disabilities [23]. Therefore, the SVM has been selected as one of the classification methods in this implementation of ITS (MIRACLE). In a study, learning disabilities are predicted using rough set theory, emphasising

data mining strategies for more accurate results [24]. Artificial Neural Network (ANN) has been implemented in a study for the disability's classification purpose. The single input layer in this ANN model depicts the different symptoms at the birth of a child, and maternal health and output represent several types of disabilities [26]. A new algorithm has been proposed in a study based on correlation for missing values and reduced the redundant values from the data for an effective learning disability identification process. The developed missing value imputation model has been illustrated as an effective prediction system [25].

3 Proposed framework

This study is focused on designing and developing a learner model and providing a personalised learner profile adapted to the learning difficulties. The schools and psychotherapeutics centres have collaborated for data collection. The team visited the targeted six schools and two psychotherapeutic centres to collect data and develop a questionnaire framework. The initial screening of the learner model for the learning disabled is proposed in two grades, grade 1 and grade 2, as depicted in Fig. 1. Irrespective of age and experience, the learner gets assigned to a particular grade based on level of knowledge. Each grade consists of 19 pretest questions which are the basis to develop learner profile with learning difficulties.

The audio data is collected to extract speech features, handwriting, and drawing samples for extracting features associated with writing issues. The speech features considered for the classification process are Phonological Awareness, Literacy Skill, Random Naming, Reading Fluency, and Loudness [27]. Motor Skills, Space Knowledge, Spellings, Copying Skill are features extracted for writing analysis. Some of the quizzes have been collected from learners as additional features. These additional features were mapped to the Cognitive Strength (CS), Learning Preferences/Style (LS), and Problematic Skills (PS) of the learners [28–30]. The quiz has generated data in terms of score and time. The intermediate

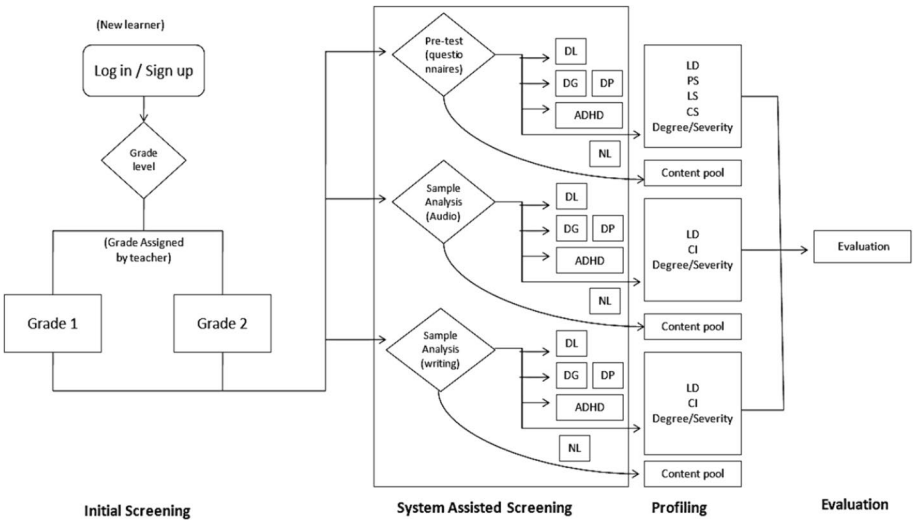


Fig. 1 Flowchart of the learner module

output, which is mapped with each pretest and sample of a questionnaire, will be considered for profile development later. These intermediate outputs are *Problematic skills, Cognitive Strength, Learning Preference/Style*. The Problematic skills will indicate the skills the person lacks in solving pretest quizzes. The Cognitive strength represents the various mental activities performed by the learner associated with the questions of learning and problem-solving. It will map the strongest cognitive strength of the learner while performing in pretest. The nine cognitive strengths are Long-term memory, Short-term memory, Attention, Concentration, Orientation, Abstraction/Rational thinking, Language Ability, Visual Perception and fluency [31]. The proposed work includes LD-specific cognitive strengths such as memory and attention, visual perception, and rational and logical thinking (processing).

The learning preference in the teaching process plays a crucial role in academics and learning outcomes. The author of many studies has included Felder- Silverman learning styles model in educational and tutoring models for adaptive systems [29]. In this work, we have framed out four learning styles reflective, visual, active, and kinaesthetic. In order to deliver the content according to the learning interest of the learners, a learning style along with cognitive abilities is required. The studies have shown the positive impact of using more than one learning style in the learning process. The known learning models are of Klobe Model (Active/Reflective/Observant/Experience), Felder Silverman Model (Active/Reflective, Sensory/Intuitive, Visual/Verbal, Sequential/Global), and VARK Model (Visual, Aural, Read, Kinaesthetic) [32]. The learning model of Klobe, Felder-Silverman, and Vark have considered four types of a learner as Reflective (Klobe), Visual (Felder Silverman), Active (Felder Silverman), and Kinaesthetic (Vark). In the present study, LD-specific learning preferences and styles have been classified into story-based learning (slides), video-based learning, game-based learning, and practice sets.

The final profile will include the details about LDs or LDs Type, category, cognitive strength, learning style, and problematic skills, the process flow of the proposed ITS is given in Fig. 1.

3.1 Methods

3.1.1 Participants and inclusion/exclusion criterion

The sample of 304 learners has been selected, comprising twenty-two dyslexic learners, twenty-nine learners with Dyslexia and Dysgraphia difficulty, and seven with all Dyslexia, Dysgraphia, and Dyscalculia difficulties. All the rest learners were non-learning learners with disability. The learners were either from the Doon psychotherapeutic centre or registered at some government and public schools in the Dehradun district of Uttarakhand state of India. The data was collected and stored between March 2020 and June 2021 from these centres and schools. The average participant's ages were between 7–12 years. They were interviewed for the pretest and created a learner database. A pretest was conducted with 19 questionnaires to identify learning disabilities Dyslexia, Dysgraphia and Dyscalculia. These questionnaires (Please refer Appendix 1) were included in this experiment and learners were interviewed face to face for pre-test and post-test data collection.

The participation of learners was based on the following selection criterion.

1. The teacher or parent should validate the candidate's academic performance.
2. The participants should not have any visual, hearing, cognitive, or speech impairment.

3. Any participant should not have any linguistic barrier and be familiar with English and Hindi languages.

Classifier Module LSTM, SVM, and AHP are integrated for the proposed module for LDs identification and profile generation. The LSTM and SVM have already been utilised to predict diseases in clinical research. The involved approaches and the model implementation and evaluation process has been discussed under this section.

3.2 Long-Short Term Memory (LSTM)

The attributes employed in discriminating and identifying Dyslexia, Dysgraphia, and Dyscalculia are directly linked to the specific challenges associated with each condition. These challenges encompass disruptions in reading skills, fluency, writing abilities, and reasoning for the affected learners. For Dyslexia, the learners' speech audios were gathered and processed utilizing LSTM technique to classify speech features, including Phonological Awareness, Literacy, Random Naming, fluency, and Loudness. The LSTM with the same structure as RNN has four neurons interacting altogether. LSTM is used for time series data that perform well for speech and image recognition processes. The LSTM approach involves extensive calculations in the gates, calculating the number of features in the concurrent neural network. All the calculations generated through the gates are based on the given equations from 1 to 7. The LSTM model passes the information by the gate structure to update the memory cells state, given as Eq. 4. The gate structure includes the input, given as Eq. 2, Forget gate is depicted through Eq. 1, and Output gate calculations are represented as Eq. 5. The Memory cells have sigmoid layers and also include the tanh layer. The Forgotten state gate given as Eq. 1 tells the state which is no longer used in the model. The LSTM model consists of the following prime components – Memory cell, Forget cell, Input gate, and Output gate. Figure 2 represents the LSTM structure with the associated gates.

$$f_t = \sigma(w_f.\left[h_{t-1}, x_t\right] + b_f) \text{ (forget gate)} \tag{1}$$

$$i_t = \sigma(w_i.\left[h_{t-1}, x_t\right] + b_i) \text{ (input gate)} \tag{2}$$

$$\bar{z}_t = tanh(w_z\left[h_{t-1}, x_t\right] + b_z) \text{ (block input)} \tag{3}$$

$$z_t = f_t * z_{t-1} + i_t * \bar{z}) \text{ (new cell state)} \tag{4}$$

$$o_t = \sigma(w_o.\left[h_{t-1}, x_t\right] + b_o) \text{ (output gate)} \tag{5}$$

$$h_t = o_t * \tanh(C_t) \text{ (block output)} \tag{6}$$

Here, in the equations described above σ is the sigmoid function to decide which information to keep or discard. The i_t, f_t,$o_t \epsilon$ $[0, 1]^n$, z_{t-1} is old cell state and z_t is the new cell state. The x^t denotes the input vector at time *t, W* are rectangular input weight matrices,

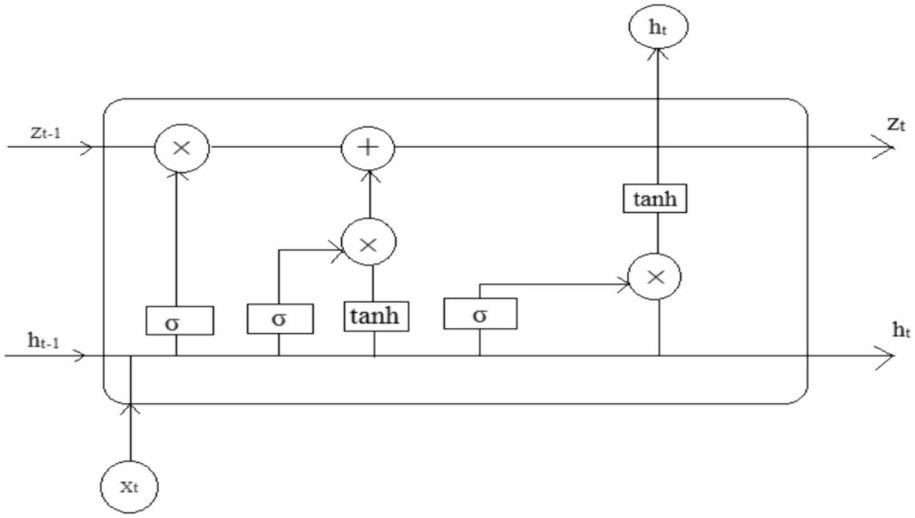


Fig. 2 Structure of Long short term memory (LSTM)

b are bias vectors, *tanh* is the hyperbolic tangent activation function. Forget gate update the auxiliary information, eliminating vanishing and exploding gradient problem.

The output layer is configured as a *softmax layer,* given as Eq. 7, where hidden units map input x_j to a class probability p_j in the form

$$Pj = \frac{exp(xj)}{\sum i(xi)} \tag{7}$$

As a cost function for backpropagating gradients in the training stage, we use the cross-entropy function defined in Eq. 8.

$$C = \sum_{j} tjlogpj \tag{8}$$

Where, t_j represents the target probability of the class j for the current evaluated example, taking a value of either 1 (true class) or 0 (other class). In LSTM, a large number of calculations are performed in the gates. The weight matrix dimension input by the input gate, the output gate, and the cell state, and the weight matrix dimension at the last time, and the output matrix dimension connected to the output of the neural network, where and are the dimensions of the input and the output, respectively, and is the number of memory cells.

3.3 Support Vector Machine (SVM)

In the present study SVM, classifies Dysgraphia and Dyscalculia features such as motor skills, spatial knowledge, spellings, legibility, visual-spatial response, sentence word structure, reasoning and word formation. SVM, which is utilized extensively for classification, regression, and density estimation. SVM methodology creates a hyperplane w. $x + b = 0$,

$x_{i\,\epsilon}R.^{n}$ to segregate the data elements x_i (the data elements that resides on the same side of the generated hyperplane) that correspond to a given decision $g(x) = sign(w.x + b)$

In the hyperplane generation task, the distance to the closest data element is maximized to minimize the risk of incorrectly classifying the raw data. The SVM approach aims to find out the classifier of the form given in Eq. 9:

$$y(x) = sign\left[\sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b\right] \tag{9}$$

$\alpha_{i,}$are the positive real constants,
b, is the real constant,
$K(x_i, x)$, represents the inner product operation,
Figure 3 represents a separating hyperplane for a training set.

Classification models have been implemented using Python's scikit-learn package (version 0.21) (version 3.6.3). Each classification method is compared with all classification model performances. Additionally, the features considered for the classification process were extended to include cognitive aspects. Responses obtained from the learners' questionnaires were integrated to assess their cognitive strengths and learning preferences. This assessment was conducted using the Analytic Hierarchy Process (AHP) technique. This comprehensive approach allows for a thorough examination of both the linguistic and cognitive dimensions, enhancing the discrimination and identification of these learning challenges.

3.4 Analytical Hierarchical Process (AHP)

The Analytical Hierarchical Process (AHP) is developed to generate decisions based on multiple criteria. Satty, in the 1980s, has proposed a strong and helpful mathematical tool

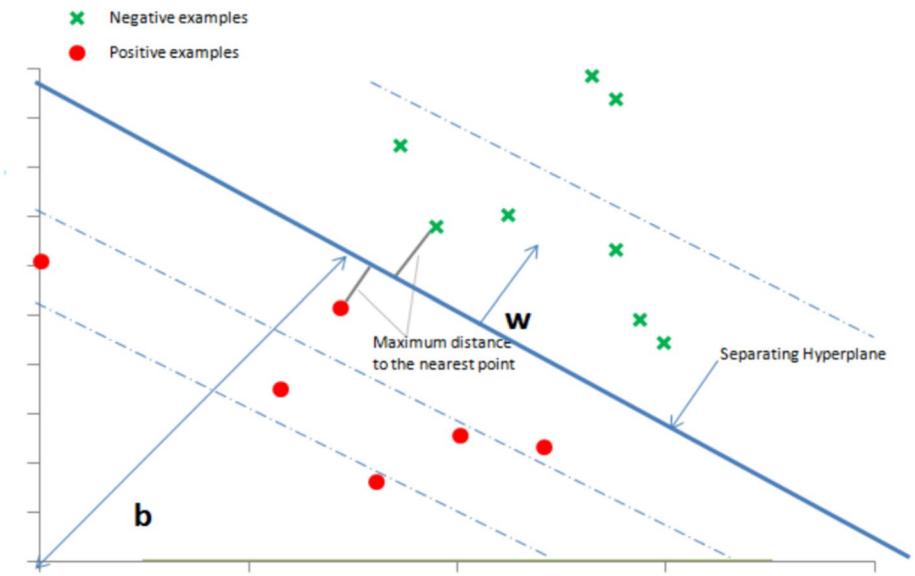


Fig. 3 Representation of a separating hyperplane for a 2-D training set

for the management of qualitative and quantitative criterion elements involving the decision process [33]. The first step in AHP is to construct a hierarchy of the problem; the second is to assign the nominal value to each level of this hierarchy and create a matrix for the pair-wise comparison judgment. There are three steps in the AHP model methodology: structuring the problem, paired comparison, and priority analysis.

3.4.1 AHP process for ranking the Learning disorders

Step1 Structuring the problem

At the top level of the hierarchy is considered level1 and is the objective of the model's decision process. Directly below is level 2 of the model, related to level 1 of the model. In level 2, the criteria are essential in the learning disability ranking process. The criteria that influence the overall AHP decision process model are Reflective learning, Visual learning, and Active Learning (as discussed above). For clarity, the evaluation of these criteria is based on the more important criteria. In level 3, the sub-criteria are used as a direct reference for evaluating the defined decision alternatives. The sub-criteria in level 3 is utilised as the base for evaluating each decision alternative, as given in Fig. 4. For this study, specific learning preferences/styles for learners with learning disabilities (LD) have been identified, including story-based learning (slides), video-based learning, game-based learning, and practice sets.

Once all the criteria and sub-criteria are structured according to the hierarchical tree structure, the second phase is a pair-wise comparison of these predefined criteria of the structured tree.

Step2 Paired Comparisons

In this stage of the AHP mathematical model, the paired-wise comparison will be conducted within the sub-criteria of the same level. The pre-judgment scale is used for

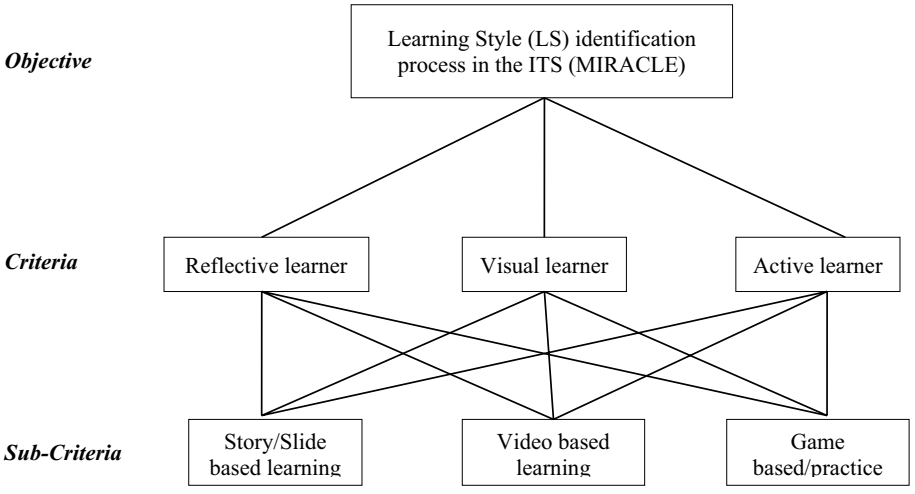


Fig. 4 The hierarchical structure of the criteria

comparison and to determine the preference of each element. The comparison matrix generated for the sub-criteria: Dyslexia, Dysgraphia and Dyscalculia as shown in Tables 1, 2 and 3.

Table 1 represents the pairwise comparisons for the Reflective sub-criteria (C1)—Story/slides, Video based, and Game/practice set. The values in the matrices indicate the strength of preference or importance, and the comparisons are done with respect to a certain criterion.

Story/slides to Video based: Story/slides is considered to be 5 times more important than Video based.

Video based to Game/practice set: Video based is considered to be 0.2 times as important as Game/practice set.

Story/slides to Game/practice set: Story/slides is considered to be 2 times more important than Game/practice set.

This Table 2 represents the pairwise comparisons for the Reflective sub-criteria (C2).

Video based to Story/slides: Video based is considered to be 6 times more important than Story/slides.

Game/practice set to Story/slides: Game/practice set is considered to be 7 times more important than Story/slides.

Game/practice set to Video based: Game/practice set is considered to be 0.2 times as important as Video based.

This Table 3 represents the pairwise comparisons for the Active sub-criteria (C3).

Story/slides to Video based: Story/slides is considered to be 4 times more important than Video based.

Table 1 The Comparison Matrix for the Reflective sub-criteria (C1)

Criteria(C1)	Story/slides	Video based	Game/ practice set
Story/slides	1.00	0.20	2.00
Video based	5.00	1.00	5.00
Game/practice set	0.50	0.20	1.00

Table 2 The Comparison Matrix for the Reflective sub-criteria (C2)

Criteria(C2)	Story/slides	Video based	Game/ practice set
Story/slides	1.00	6.00	7.00
Video based	0.16	1.00	5.00
Game/practice set	0.14	0.20	1.00

Table 3 The Comparison Matrix for the Active sub-criteria (C3)

Criteria(C3)	Story/slides	Video based	Game/ practice set
Story/slides	1.00	0.25	0.12
Video based	5.00	1.00	0.16
Game/practice set	8.00	6.00	1.00

Video based to Game/practice set: Video based is considered to be 0.16 times as important as Game/practice set.

Story/slides to Game/practice set: Story/slides is considered to be 8 times more important than Game/practice set.

Step3 Priority Analysis

In this step of the AHP process, these comparisons are standardised after the paired-wise comparison has been constructed. The one standardisation process which is frequently used is to divide each number in a column of the paired wise comparison table with the total sum value of the particular column. Afterward, the arithmetic mean of each line is calculated for the standardised matrix, and the respective priorities for each criterion are obtained. Table 4 represents the matrix for the Reading Skill sub-criteria in % (Tables 5 and 6).

3.5 Model development

The two cascading models in which one LSTM performs the data reduction part and SVM classifies the learning disabilities. The data predicted correctly by the first model is used as input in the second model for later classification. Moreover, AHP has been incorporated with this cascading model for the additional profile details such as Learning Style (LS),

Table 4 The Comparison Matrix for the Reading skill sub-criteria in %

Criteria(C1)	Story/slides	Video based	Game/practice set	Priority (%)
Story/slides	0.153	0.142	0.250	18.223
Video based	0.769	0.714	0.625	70.283
Game/practice set	0.076	0.142	0.125	11.492
Sum	1	1	1	100%

Table 5 The Comparison Matrix for the Writing skill sub-criteria

Criteria(C2)	Story/slides	Video based	Game/practice set	Priority (%)
Story/slides	0.763	0.833	0.538	71.181
Video based	0.127	0.138	0.384	21.692
Game/practice set	0.109	0.027	0.076	7.126
Sum	1	1	1	100%

Table 6 The Comparison Matrix for the Basic Math skill sub-criteria

Criteria(C2)	Story/slides	Video based	Game/practice set	Priority (%)
Story/slides	0.071	0.034	0.096	6.756
Video based	0.357	0.137	0.129	20.803
Game/practice set	0.571	0.827	0.774	72.440
Sum	1	1	1	100%

Cognitive Strength (CS), and Problematic Skills (PS). The AHP model has been selected for the optimal selection from the presented Learning styles, Cognitive strengths, and Problematic skills as the intermediate output. Figure 5 demonstrates the combination of the overall model process.

3.5.1 SVM, LSTM and AHP Integration within the ITS framework

Support Vector Machine (SVM), Long Short-Term Memory (LSTM), and Analytic Hierarchy Process (AHP) Integration within the framework involves orchestrating these models for a cohesive and comprehensive learning disability classification system. The steps involved in the process are:

Data Preprocessing: Speech and writing features were preprocessed using advanced techniques for high-quality data analysis. Specifically, Mel-Frequency Cepstral Coefficients (MFCC) were used for speech feature extraction [34], and the Histogram of Oriented Gradients (HOG) technique was applied for writing feature extraction [35]. These methods provide robust feature representations, ideal for subsequent classification using Long Short-Term Memory (LSTM) networks. The LSTM classified audio data from surveys, distinguishing between "Learning Disabled" and "Non-Learning Disabled," with instances labeled as "Non-Learning Disabled" excluded from the final analysis.
Analytical Hierarchy Process (AHP) Integration: Apply AHP to evaluate factors related to learning disabilities and assign weights within the defined hierarchy. Conduct comparisons among alternatives (Dyslexia, Dysgraphia, Dyscalculia) using AHP, calculating probabilities for each alternative's impact on learning outcomes.
Learning Disability Ranking: Rank disabilities based on Severity, Learning Preferences/Style, Cognitive Strength, and Problematic Skills using AHP-calculated probabilities.
Framework Integration: Integrate SVM, LSTM, and AHP models into the ITS framework. Design ITS module, ensuring fluid data transitions.
Classification: Execute the classification task on the reduced dataset using the integrated SVM model. Assign learning or non-learning-disabled labels using SVM method and determine the disability category/severity based on AHP rankings.

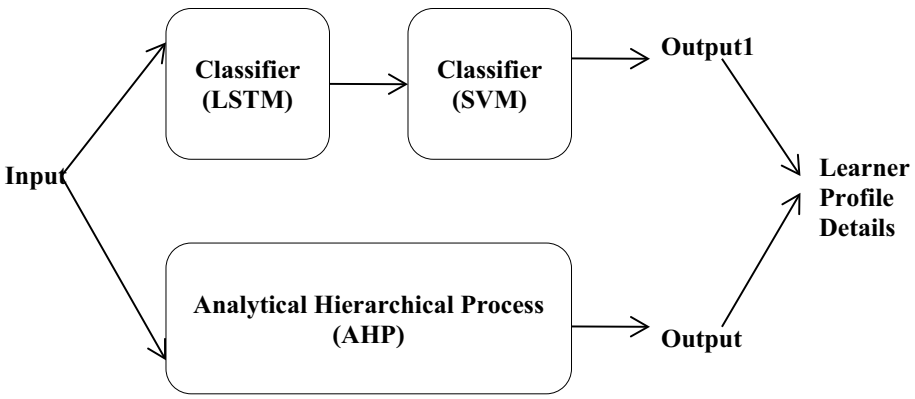


Fig. 5 A classifier combined with another classifier with intermediate AHP output

The study is based on cascading the LSTM and SVM learning model; the former in this module will reduce the audio data collected through surveys of children (with or without) learning disabilities. The LSTM generates classification results under two classification labels, "Learning Disabled" or "Non- Learning Disabled" all data with the label "Non-learning disabled" will then be eliminated from the database. Here, AHP will eliminate the incorrect and inconsistent leaner data from the dataset. The AHP transforms the comparison; the weight of each factor related to the disability allows the assessment of each feature inside the defined hierarchy. Further, a comparison is being made, the probability related to each alternative (Dyslexia, Dysgraphia, Dyscalculia) will be calculated. This probability determines the possibility of the alternative to affect the learning outcomes of the learning disabled. If higher is the probability, higher is the rank of the predefined alternative. This assessment will rank the disability in terms of its Severity, Learning Preferences/Style (LS), Cognitive Strength (CS), and Problematic Skills (PS). The final task is to classify the reduced dataset into learning or non-learning-disabled labels and the category/severity of learning disability that the integrated AHP model has already ranked.

4 Result and discussion

Research Question 1: To design and develop an Intelligent Tutoring System for the learning disabled (MIRACLE) for learning disabilities prediction, profile, and content generation

The ITS (MIRACLE) has been implemented and integrated utilising Long Short Term Memory (LSTM), Support Vector Machine (SVM), and Analytical Hierarchy Process (AHP) techniques. The ITS (MIRACLE) for the LDs is based on the cascading model architecture where the output from one learning model has served as the input to another model. This has effectively reduced the cluttered data at the beginning of the module development. Moreover, the AHP mathematical model has added the benefits of ranking the disabilities in terms of *Severity, Learning Style, cognitive strength, and Problematic Skills* of the learner. Figure 6 shows the developed ITS (MIRACLE) for LDs. Figure 7 represents the learner details obtained from the ITS (MIRACLE).

This profiling system is dynamic and presents detailed information expected from an ITS system. The proposed model provides the following advantages over the existed models:

1. De-cluttered the ambiguous and unwanted data at the initial stage through LSTM classification process. This has improved the data quality significantly for further classification.
2. Incorporation of the AHP process has added the advantage of a detailed learner profile consisting of LD with category, LD subtype, Learning Style (LS), Cognitive Strength (CS), Problematic Skill (PS).
3. This ITS (MIRACLE) can present learner-specific content according to the generated learner profile details.
4. The performance of this proposed model in the detailed learner profiling task of Learning Disabled is significantly effective.

Research Question 2: ITS (MIRACLE) Evaluation

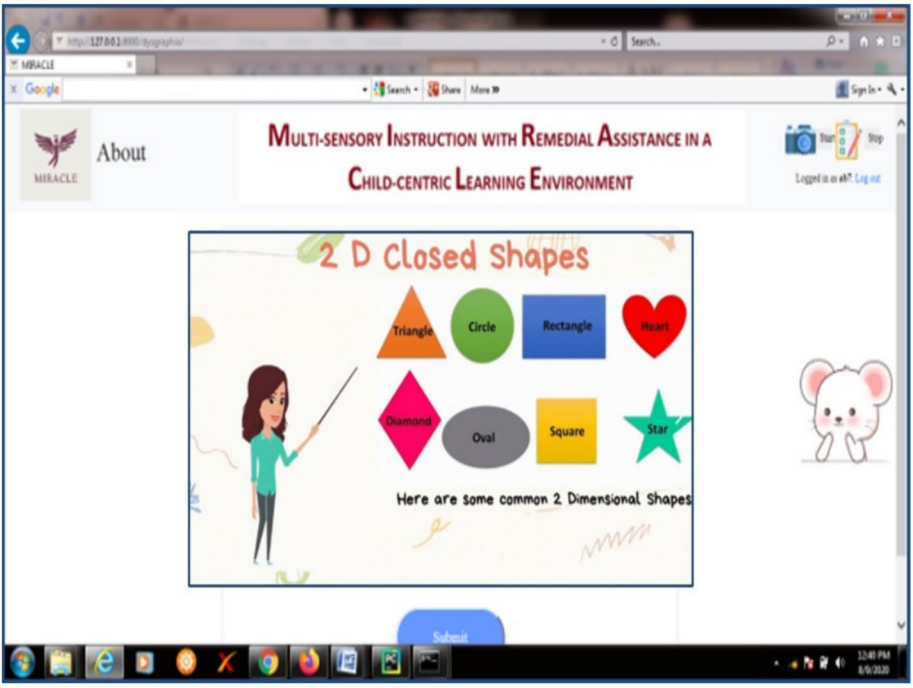


Fig. 6 ITS (MIRACLE) for LDs

The features selected from the dataset include questionnaires related to Literacy skills, Phonological Awareness, Word Formation, Sentence Structure, Counting Numbers, Calculation, Reasoning, Visual-spatial Response, Rhyming, Numeracy skills, Handwriting Skills, Word Fluency, Copying Skills, Loudness. These data points are trained over the proposed model.

4.1 Performance evaluation of the proposed model

Figure 8 depicts the values of evaluation metrics obtained in this experiment. The data is trained and tested with a ratio of 8:2. The results obtained through the proposed model are compared with the output of the similar study conducted by ANN, SVM, and LSTM on evaluation metrics of precision, recall, accuracy, and F-1 score. The combined model LSTM and SVM evaluation is based on evaluation matrices Precision, Recall, Accuracy, and F1 score. Moreover, the AHP features have been combined to evaluate the learner's performance to present the optimal selected profile content and other required details. The classification model has been selected based on its comparison with other machine learning models on performance metrics (Accuracy, Precision, Recall, and F1-score). The accuracy is used as a performance indicator given as Eq. 1, true positive- correctly predicted positive samples and false positive-incorrectly predicted positive samples, false negative- incorrectly predicted negative samples, and true negative- correctly predicted negative instances are used to calculate these performance indicators.



Fig. 7 Profile details

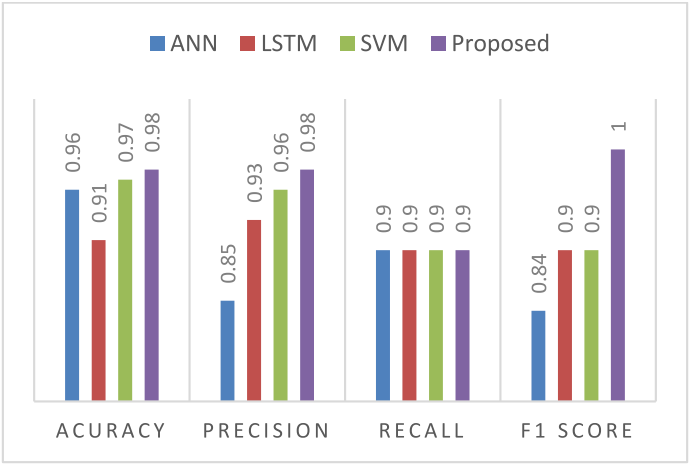


Fig. 8 Algorithm Comparison

$$Accuracy = \frac{\text{True Positive} + \text{True Negative}}{(\text{True Positive} + \text{False Positive} + \text{True Negative} + \text{False Negative})} \tag{10}$$

Precision is the total positive predicted samples out of the predicted samples; Eq. 2 is the precision formula.

$$Precision(p) = \frac{\text{True Positive}}{(\text{True Positive} + \text{False Negative})} \tag{11}$$

Recall is the fraction of the positive samples that are correctly classified; the formula is given in Eq. 3.

$$Recall(r) = \frac{\text{True Positive}}{(\text{True Positive} + \text{True Negative})} \tag{12}$$

The metric F-1 score represents the harmonic mean of the recall and precision values. Equation 4 represents F- measure metric.

$$F1score = \frac{2 * \text{p} * \text{r}}{p + r} \tag{13}$$

The proposed Model for LDs identification and discrimination from other disabilities generates promising results. The precision obtained of ANN, LSTM, SVM, and the proposed model is approximately 85%, 93%, 96%, and 98%, respectively. The accuracy of the proposed model is the highest, with a 98% score value. The algorithms, ANN, LSTM, and SVM have obtained 96%, 91%, and 97% accuracy. The F-1 score of these algorithms is compared, and the proposed algorithm yielded the highest among the entire algorithm's scores. Therefore, it is evident from the results that the proposed model has outperformed other algorithms in this experiment.

4.2 Pretest and Post-test evaluation of the proposed ITS (MIRACLE)

In Fig. 9, the pretest without utilizing proposed ITS to identify a learner as LD (Learning Disabled) or NLD (Non-Learning Disabled) is quite significant but not influential compared with the proposed model post-test results. There are improvements in classifications values shown as the result line of the post-test using the proposed model. The report generated using the proposed model can categorise the profile of learners with disability into the respective category along with their Learning Style/Preference (LS), Cognitive Strength (CS), and related Problematic Skills (PS) because the Proposed Model classification for learner profiling provides category range (Extreme, High, Low, and Nil). The category Nil represents the learner who does not have symptoms of the related disability; the severity of the learning difficulties decreases from extreme to nil categories in a profile. The output range is precise and flexible (can be extended or reduced) according to need.

ANOVA (One-Way) test is applied on the data obtained through the proposed Learner module in the ITS of the LD. The ANOVA (One-Way) is conducted on pretest result of the learner module in ITS assuming the samples have equal variance. The ANOVA (One-Way) is selected for evaluation as more than two samples results of the population was used in the pre-test for analysis.

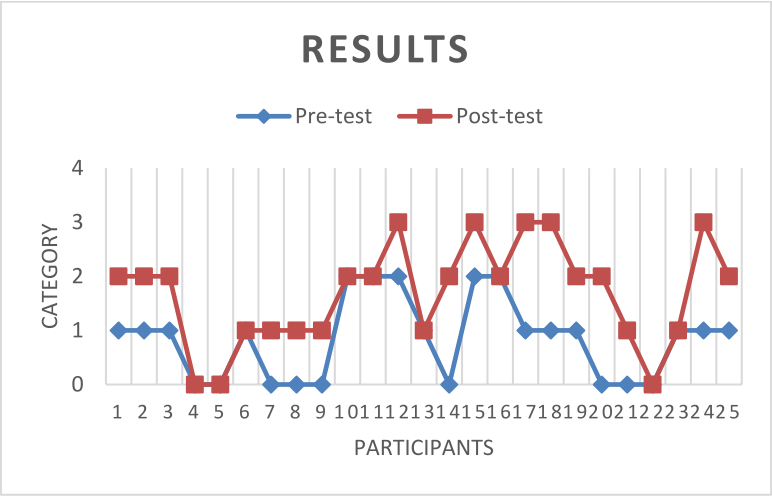


Fig. 9 Pretest and post-test analysis graph of implemented the ITS

Hypothesis There is no significant difference present in the mean of each sample, in the learner module of the implemented ITS of LD.

In Table 7 the alpha value (0.05) is more than P value (0.006108), so the null hypothesis remains unaccepted. Hence, it is concluded that the implemented algorithms for the development of the learner module in the ITS of LDs identification was significant and improves learner specific profiling. Therefore, the proposed method to create learner module is significant and difference in effectiveness found in the methods utilized.

However, it is unknown that which of the specific sample differed and is significant. The Posthoc test is generally preferred test to determine the differed sample group on ANOVA (One-Way). Table 8 is the post hoc test analysis and the alpha-value obtained in post hoc test is (0.0125). The alpha value when compared with the P-value obtained in the post hoc test the difference between the proposed sample group mean is found to be statistically significant. This post hoc analysis revealed that the proposed method for learner module development in the ITS of LD has obtained significant results.

The above test results proved that the learning outcomes in Learner Model has significantly improved than the existing ITS Learner Models. This improvement in the learning is due to the integration of LSTM and AHP modelling introduced in the Learner Model into the ITS. Hence, it is concluded that integrating the proposed Learner Modelling methodology in the ITS model is significant and improves learning for LD and NLD learner.

Table 7 ANOVA test result

ANOVA						
Source of Variation	*SS*	*Df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	3.888	3	1.296	4.174	0.006	2.704
Within Groups	28.252	91	0.310			
Total	32.141	94				

Table 8 POST HOC test

POST-HOC TEST		
Groups	*P-value(T-test)*	*Significant (Yes/No)*
Proposed ITS v AR based ITS [36]	0.105	No
DALI-DAB v Proposed ITS [37]	0.209	No
Proposed ITS v FMMN architecture [38]	0.001	Yes
ITS with Instructional framework for LD v Proposed ITS [39]	0.002	Yes

The main contribution of this study is the development of an ITS (MIRACLE) for learning disabilities identification and learner profile generation. Moreover, based on the generated learner profile, the content is generated for educated learners. The limitation of the proposed work is:

1. A limited amount of data has been utilised to develop the proposed model. However, the data would increase quality and quantity in future studies.
2. The research focused on a specific context or educational environment. It is important to acknowledge that the effectiveness of the MIRACLE framework might vary when applied to different educational contexts, institutions, or countries.
3. The identification of learning disabilities was based on a pretest and non-invasive tools. The accuracy and validity of these tools in capturing the full spectrum of learning disabilities may be a limitation, as the diagnosis of learning disabilities often requires comprehensive assessments.
4. The success of the MIRACLE framework heavily relies on the chosen machine learning models (LSTM, SVM, and AHP). Limitations or biases inherent in these models may impact the accuracy of learning disability identification.

5 Conclusion and future direction

This study has implemented an ITS (MIRACLE) to profile learners with disability and generate learner-specific content. The study classified learners with disability and their learning preferences, cognitive strength, and problematic skills as profile details. Moreover, in this study, the results in profile are improved by classifying the subtype of the learning disabilities. This framework is a cascading module developed combining the advantages from machine learning models (LSTM + SVM) and mathematical model Analytical Hierarchical Process (AHP). This module can outperform the other implemented architectures for the learning disabled and proves to be a better profiling system. In the future, studies will improve the existing ITS (MIRACLE) performance by incorporating the EEG device dataset for LDs profiling. The data has a significant role in improving the overall accuracies of the developed model. Thus, the focus is to improve the existing dataset to increase the ITS (MIRACLE) efficacy. Additionally, the proposed ITS can be integrated into vocational training programs to simulate real-world scenarios, enhancing practical skills. This could

facilitate flexible learning schedules, allowing students to improve their learning outcomes. It provides personalized support for individuals with varying levels of prior knowledge.

Appendix 1: A set of Pre-test Questions

Quiz number Pre-test Questions

Grade 1

1 Click on the letter matching given letter: V
2 Click on the letter matching given letter: B
3 Listen to audio and fill in the blanks: C P
4 Arrange the letter to form a Word shown in the picture: L T I L T E / Y L O E L W / D C U K
5 Which picture best describes the word: PLAY
6 Count the fish and click the number
7 ADD:
  5 + 3
8 Subtract:
  8–2
9 Write the missing number: 5, 7,,11
10 Click on the left picture:
11 a. Say CUPCAKE without CAKE
  b. Say MATCHBOX without BOX
12 a. Count backwards from: 30–21
  b. True/False: 2 > 3
  c. Write in words: 97
  d. Write the digits shown: 5 2 2 7
  e. Subtract: 28–2
  f. Addition: 28 + 2
13 Connect the dotted fish imageand upload.
14 a. Pronounce the words given correctly:af am ab zikzab at am of see look
  b. Name the colors shown below?
15 Copy the lines in.txt file and upload
16 a. A bee has 6 legs. How many legs do 2 bees have?
  b. You have 7 cookies and you ate 2 of them. How many cookies do you have left?
17 Write the given line on a blank sheet, scan and upload.
18 a. Do they rhyme? HIT and HEIGHT say yes or no
  b. Do they rhyme? CUP and PUP say yes or no
  c. Merge the words PENCIL
19 Read the passage below aloud:

Grade 2

1 1 Click on the tab matching given letters: Bfn
2 Which word rhyme with the word: FISH
3 Listen to audio and fill in the blanks:

4 Arrange the letter to form a Word shown in the picture: (Hint: I am the national flower of India.)
5 Arrange the words to form a meaningful sentence: RED ROSES ARE
6 What is the value of four in the number? 890,465
7 Solve the given problem: 23 + 17–19 = ?
8 Solve the given problem: 19*4 = ?
9 Divide: 333 by 9
10 Which dinosaur is in cylinder?
11 a. Say Confront without Con
b. Pronounce and Write INTERSTELLAR
c. Say Trustworthy without worthy
12 a. Count Backwards from: 130 to 120
b. True/False: 2550 > 2569
c. Write in words: 997
d. Write the digits shown: 6 5 5 2 2 7
e. Subtract: 528–72
f. Addition: 828 + 28
13 Help the bee to reach flowers and upload image:
14 a. Pronounce the words given correctly: zikzabziszobzam see look monkey through dark
b. Name the colours shown below?
15 Type the given text in.txt file and upload
16 a. Ellen had 380 eggs, but she lost 57 of them. How many eggs does she have now?
b. Arthur baked 35 muffins. How many more muffins does Arthur have to bake to have 83 muffins?
17 Write the given line on a blank sheet, scan and upload.
18 a. Do they rhyme? Six and Sticks say yes or no
a. Do they rhyme? Seven and Heaven say yes or no
b. Merge the words Net,working
19 Read the passage below aloud:

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Author contributions N.J.A conceived the experiments, designed the research strategy. S.D performed data collection, data analysis and wrote the manuscript with input from all authors.

Data availability Relevant key findings of this study are available within the article. All raw data generated during the current study are available from the corresponding authors upon request.

Declarations

Conflicts of interest None.

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